Taichi Kamei

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Embedded systems & Hardware engineer who takes projects from 0 to 1 under real physical constraints

Education

4th Year Engineering Physics *Expected Graduation - Dec 2028*
University of British Columbia | Vancouver

Relevant Courses: Electric Circuit Analysis, Signals and Systems, Software Construction, Digital Systems and Microcomputers, Thermodynamics, Statistical Mechanics, Analytical Mechanics, Applied Linear Algebra

Dean's Honor List 2026

Experience

Electronics & Controls Co-op *Sept 2026 - Present*
LUNR Aerospace | Toronto

- Developing state machine in NI LabVIEW for 60kN rocket engine combustion control system, integrating multiple systems with temperature, pressure, load cell sensors and valves
- Working on a modular Thrust Vector Control (TVC) firmware framework in C++, initially using it for Simulink simulation and later ported on a flight controller hardware
- Implemented Quaternion-based PID control in Python using GNC target vector and rocket body attitude
- Modeled the transfer function from attitude PID output to TVC gimbal deflection and integrated it into the closed-loop pitch/yaw model to verify stability margins

Firmware & Hardware Engineer Co-op *May - Aug 2026*
UBC Blusson Quantum Matter Institute | Vancouver

- Initially started as a PCB validation role for the 4-channel voltage controlled biasing current source PCB used for the superconducting nanowire single photon detector (SNSPD)
- Recognized the need for firmware development, designed an active object architecture in C/C++ on ESP32 with ESP-IDF, FreeRTOS, dual-core task isolation, inter-core queue communication, and event coordination, achieving deterministic concurrent per-channel control
- Ported Analog Devices no-OS 24-bit Sigma-Delta ADC and 16-bit DAC drivers to ESP-IDF, implemented SPI platform abstraction layer with manual CS toggle for 5 SPI slaves and mixed SPI modes on a 10MHz shared bus
- Developed hysteresis V-I sweep mode with 10mV step size and steady-state biasing mode with 10nA precision
- Debugged 3.3V LDO undervoltage by oscilloscope scoping and datasheet comparison, applied hardware workaround and PCB layout revision on Altium
- Tested time domain DAC and ADC drift characteristics by logging both PCB and commercial SMU measurement for 24 hours using Raspberry Pi
- Designed 8-channels co-planar waveguide PCB on Altium for 5GHz signals at 0.8 Kelvin cryogenic condition

Projects

Drone Flight Controller & Power Board *Feb 2026 - Present*

- Designed a 4-layer flight controller PCB on KiCAD, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261 chip
- Implemented power regulation stage for 4S LiPo input with 5V/5A buck converter, 3.3V LDO for sensors and MCU, and a split ground for minimizing noise from high current 15V ESC line
- Designed a 4-layer GPS-module & magnetometer PCB fitting within 26mm by 26mm
- Redesigning flight controller PCB and power board PCB as separate boards for space optimization

Autonomous Clue Detecting Robot *Nov 2025*

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane
- Created a Qt5 controller GUI which integrates simulation controls and launch of two ROS node scripts, boosting team productivity by centralizing all controls into a single window which normally takes 4+ separate terminals

Autonomous Payload Retrieving Robot *June - Aug 2025*

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

Skills

Software	C/C++, Python, LabVIEW, Rust, Java, Assembly, Simulink, VHDL, Linux, Git, CMake
Embedded	ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, ROS, Gazebo, FPGA, I2C, SPI, UART, DAQ
Electrical	Kicad, Altium, LTSpice, Soldering, Oscilloscope, Electrometer, DMM, SMU, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper